



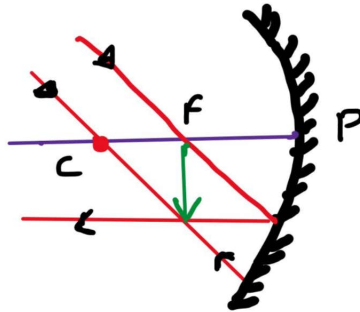
Image Formation by concave Mirror:

Position of object

Ray Diagram

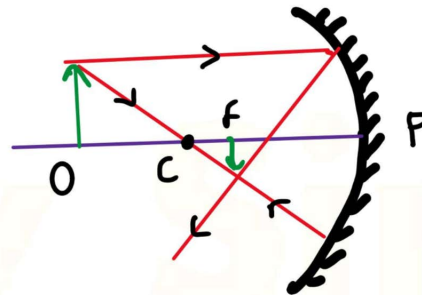
Properties of Image

At Infinity



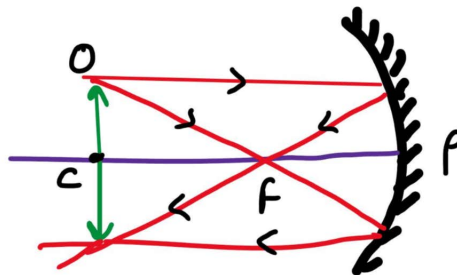
Real, Inverted,
very small at F

Between
Infinity and C



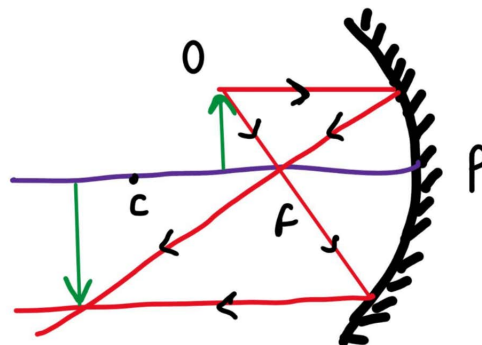
Real, Inverted,
diminished
between F & C

At C



Real, Inverted,
equal in size
at C

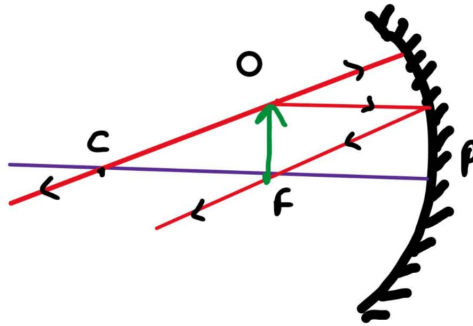
Between F
and C



Real, Inverted,
very large
between 2F and C



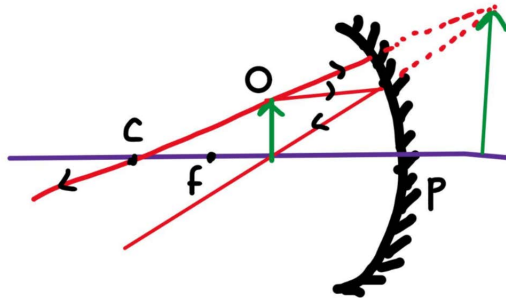
At f



Real, Inverted,
very small at f



Between f
and P



Real, Inverted,
diminished between
 f & C

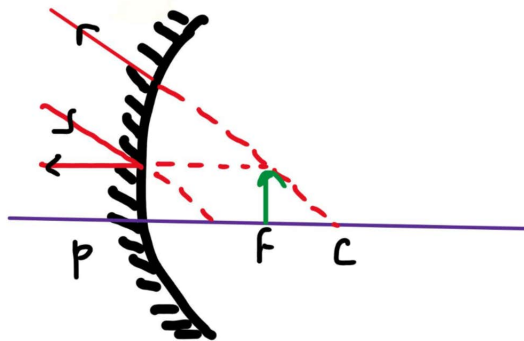
Image formation by Convex Mirror

Position of object

Ray Diagram

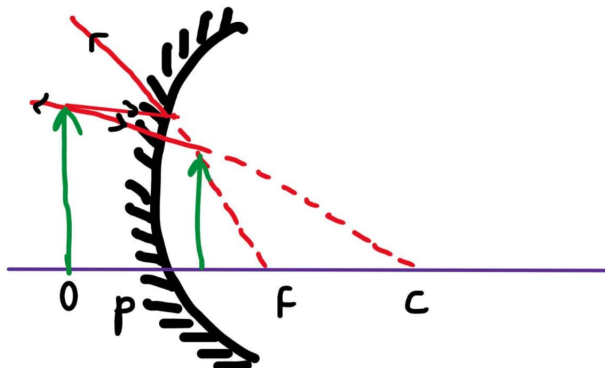
Properties of Image

At Infinity



Virtual, Erect,
very small size
at f

In front of
mirror



Virtual, erect,
diminished, b/w
 P and F





Spherical Mirrors

Magnification

$$m = \frac{h_i}{h_o} = -\frac{v}{u}$$

Motion of Images

In transverse plane

$$\vec{v}_i = m \vec{v}_o$$

(If mirror also moves)

$$v_{i/m} = m v_{o/m}$$

In longitudinal plane

$$v_i = -(m^2 v_o)$$

$$v_{i/m} = -m^2 v_{o/m}$$

Refraction of Light

Refractive Index (n)

$$n = \frac{c}{v} = \sqrt{\mu_r \epsilon_r}$$

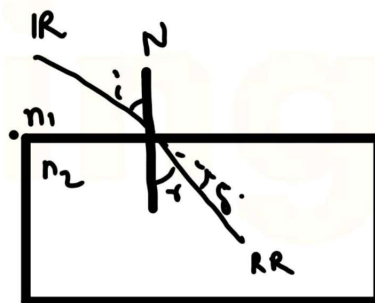
$$n = \frac{\text{speed of light in vacuum}}{\text{speed of light in medium}}$$

If $n_1 > n_2$

light travels from denser to rarer medium.

$\therefore$ light bends away from normal-

$$r > i$$



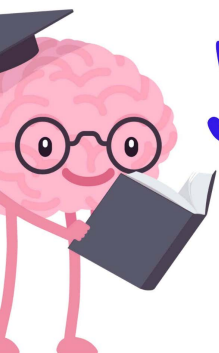
If $n_1 < n_2$

light travels from rarer to denser medium.

$\therefore$ light bends towards the normal.

$$r < i$$





For more such study
material



OUR
APP!





Important formulas

Snell's law $n_1 \sin i = n_2 \sin r$

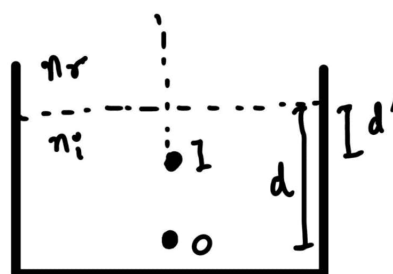
wave formula $v = f\lambda$

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

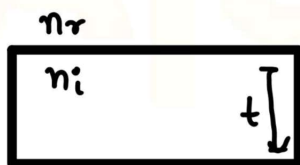
Apparent Depth

$$d' = \frac{n_r}{n_i} d$$

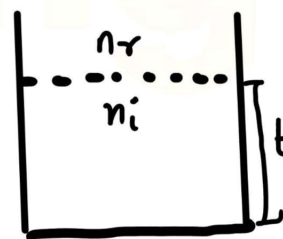
$$\text{shift} = d \left[1 - \frac{n_r}{n_i} \right]$$



Thickness of medium



$$t' = \frac{n_r}{n_i} t$$

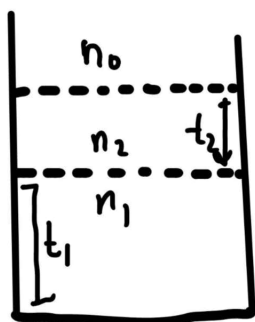


Thickness of multiple layers

$$t_1' = \frac{n_o}{n_1} t_1$$

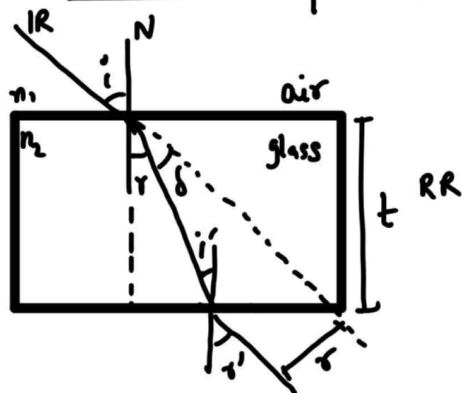
$$t_2' = \frac{n_o}{n_2} t_2$$

$$\text{Total depth} = t_1' + t_2' = \left(\frac{t_1}{n_1} + \frac{t_2}{n_2} \right) n_o$$





Lateral Displacement



$$x = t \sin i \left[1 - \frac{\cos i \sin i}{\sqrt{n^2 - \sin^2 i}} \right]$$

TIR (Total Internal Reflection)

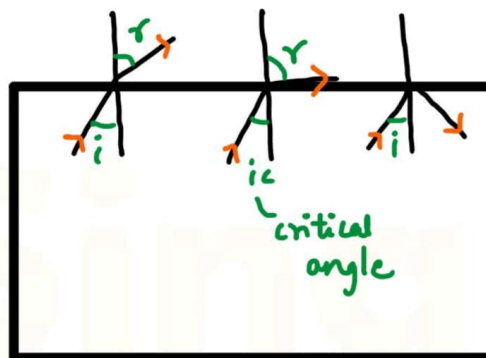
light - Denser to Rarer

$$r > i$$

$i < i_c \Rightarrow$ Refraction

$$i = i_c \Rightarrow r = 90^\circ$$

$i > i_c \Rightarrow$ TIR



Critical Angle

$$\sin i_c = \frac{n_2}{n_1}$$

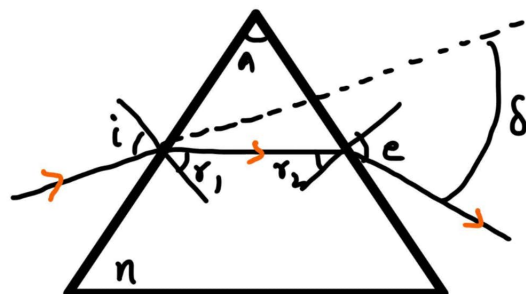
Circle of
Illuminance

$$r = h \tan i_c = \frac{n_2 h}{\sqrt{n_1^2 - n_2^2}}$$



Prism

Apply Snell's law at both surface



$$\sin i = n \sin r_1$$

$$n \sin r_2 = 1 \cdot \sin e$$

$$A = r_1 + r_2$$

$$\delta = i + e - A$$

$$\delta_{\min} = 2i - A$$

$\delta_{\min}$ when $i = e \Rightarrow r_1 = r_2$

condition for $\delta_{\min}$

$$\mu = \frac{\sin\left(\frac{\delta_{\min} + A}{2}\right)}{\sin \frac{A}{2}}$$

GRAPH

Conditions for grazing emergence

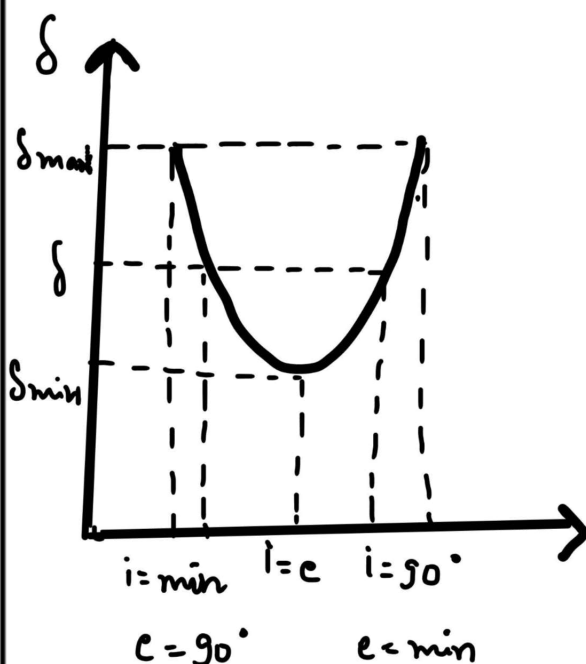
$$\sin r_2 = \frac{1}{n}$$

for equilateral and non-equilateral Prism

$$r_1 = r_2 = \frac{A}{2} = 30^\circ$$

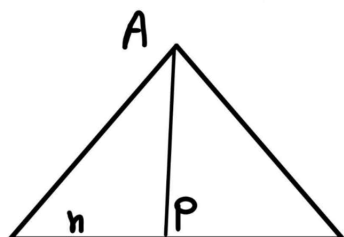
RR inside || Base of Prism

$$\angle i = \angle e$$





Time taken by prism to travel from P to A :



$$t = \frac{AP}{v} = \frac{AP}{c} n$$

for small angle prism

$$\delta = (n-1) A$$

Condition for δ_{min}

$$n \Rightarrow \frac{A + \delta_{min}}{A}$$

Refraction on Curved Surfaces

if $n_1 > n_2$ denser to rarer

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

Lateral Magnification

$$m = \frac{n_1 v}{n_2 u}$$

Principal foci

First Principal : The first principal axis from the centre of the lens

$$u = \frac{n_1 R}{n_2 - n_1}$$

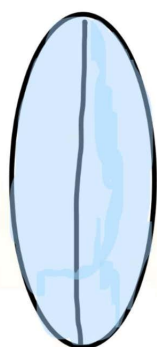


Second Principal: Position of the image on principal axis of the lens when the object is formed at infinity.

$$v = \frac{-n_2 R}{n_2 - n_1}$$

* For curved mirror first and second principal focus are same.

Lens



Bi-convex



Bi-concave

Focal length of lens

$$\frac{1}{f} = \left(\frac{n}{n_s} - 1 \right) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$n \rightarrow$ R.I of lens

$n_s \rightarrow$ R.I of surrounding

$R_1, R_2 \rightarrow$ Radius of two surfaces

for convex $\rightarrow +R$

for concave $\rightarrow -R$

Power of lens

$$P = \frac{n_s}{f}$$

$$P = (n - n_s) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$





Lens making formula
for thin lens

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Magnification

$$m = \frac{v}{u}$$

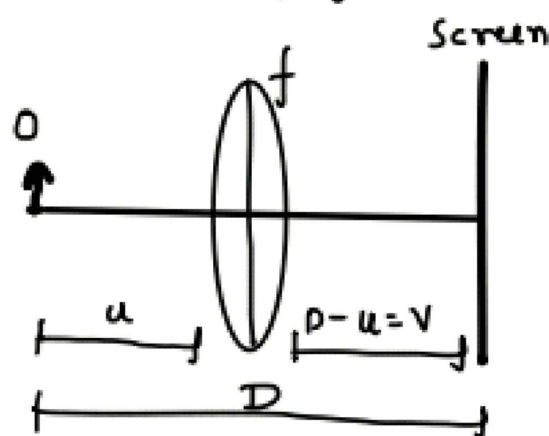
Motion

Transverse $\rightarrow \vec{v}_i = m\vec{v}_o$; Longitudinal $\rightarrow \vec{v}_i = m^2\vec{v}_o$

Motion if lens moves.

$$\vec{v}_{i/m} = m\vec{v}_{o/m} ; \vec{v}_{i/m} = m^2\vec{v}_{o/m}$$

$\Rightarrow$ Displacement method to find focal length of converging lens.



$$u = \frac{D \pm \sqrt{D^2 - 4fD}}{2}$$

If $D^2 < 4fD$

$u \rightarrow$ imaginary
i.e. no position of lens for which
a real image can be formed





$$D^2 = 4Df$$

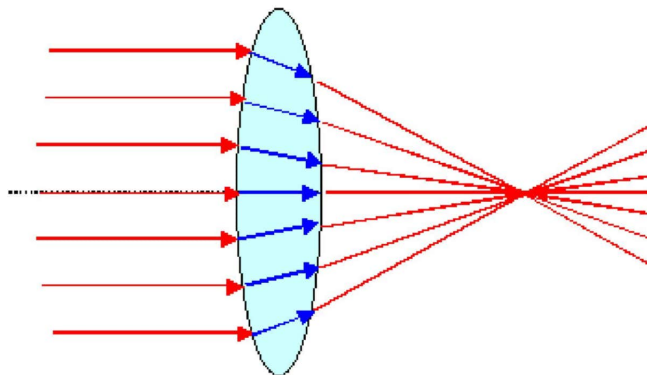
$$v = u$$

only one image is possible to form real image.

$$\text{If } D^2 > 4fD$$

Two positions are possible when real image can be formed

	$m_1 > 1$	$m_2 < 1$
u	$u_1 = \frac{D - \sqrt{D^2 - 4fD}}{2}$	$u_2 = \frac{D + \sqrt{D^2 - 4fD}}{2}$
v	$v_1 = \frac{D + \sqrt{D^2 - 4fD}}{2}$	$v_2 = \frac{D - \sqrt{D^2 - 4fD}}{2}$
m	$m_1 = \left[\frac{D + \sqrt{D^2 - 4fD}}{D - \sqrt{D^2 - 4fD}} \right]$	$m_2 = \left[\frac{D - \sqrt{D^2 - 4fD}}{D + \sqrt{D^2 - 4fD}} \right]$





Formation of Image by Convex Lens :

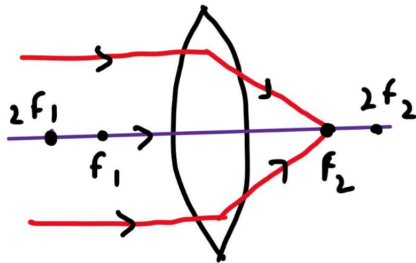
Position of Object

Ray Diagram

Position of Image

Nature & size of image

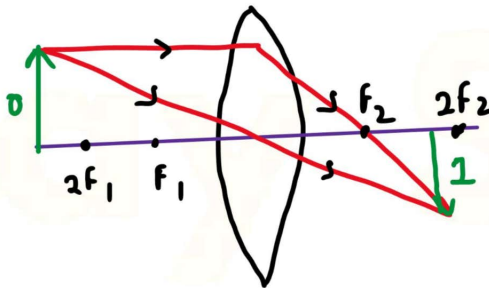
At Infinity



At the principal focus (F_2)

Real, Inverted, and highly diminished

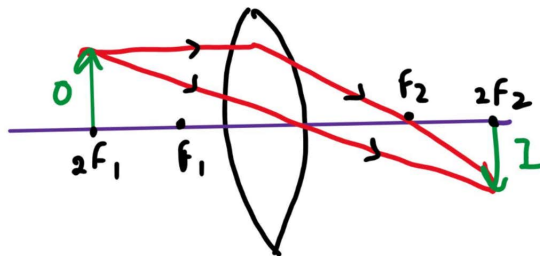
Beyond $2f_1$



Between f_2 and $2f_2$

Real, Inverted and Diminished

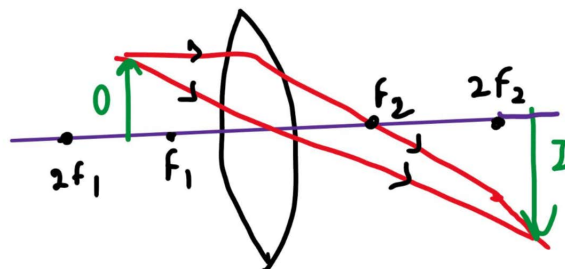
At $2f_1$



At $2f_2$

Real, Inverted and of same size as object

Between f_1 and $2f_1$



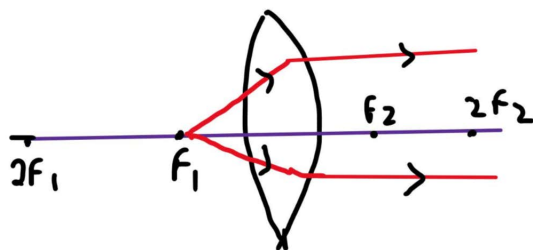
Beyond $2f_2$

Real, inverted and highly magnified





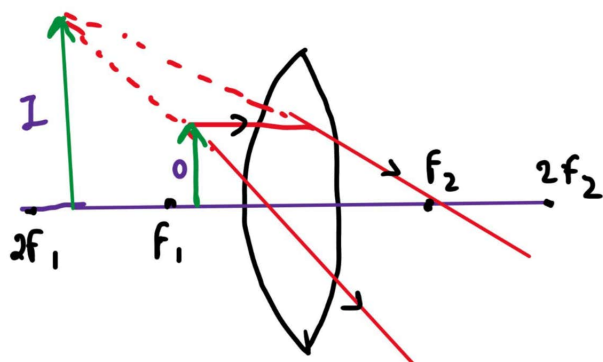
At f



At
Infinity

Real,
inverted, and
highly
magnified

Between
 f_1 and
optical
centre

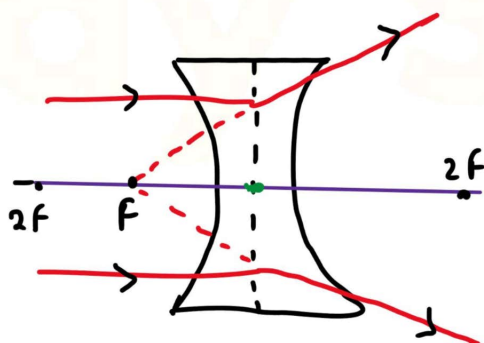


On the
same side
as the
object

Virtual,
erect and
magnified

Formation of Image by Concave Lens:

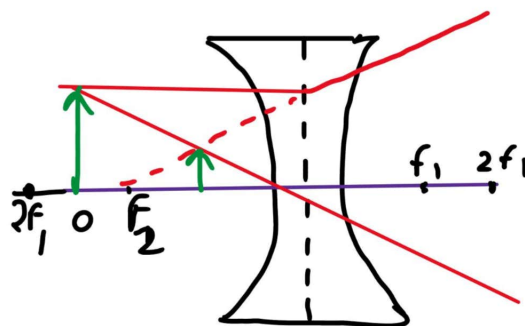
At Infinity



At the
focus

Virtual,
erect and
point size

Anywhere
except on the
principal
axis



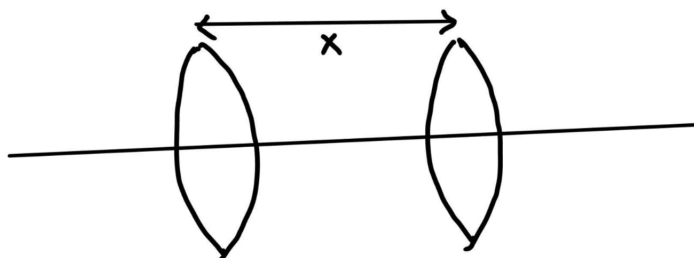
Between
lens and
 f_2

Virtual,
erect &
diminished





for two convex lenses :



$$f = \frac{x}{m_1 - m_2}$$

$$0 = \sqrt{z_1 z_2}$$

$$\frac{m_1}{m_2} = \left(\frac{D+x}{D-x} \right)^2$$

Combination of lenses:

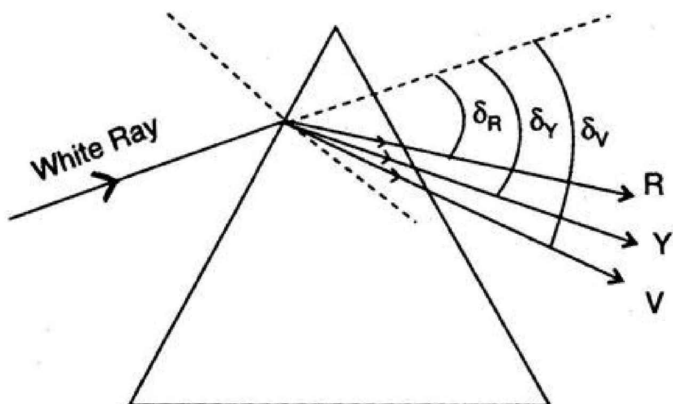
$$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} \dots$$

focal length

$$P_{eq} = P_1 + P_2 + \dots$$

Power

Dispersion of light



$$\delta = (n-1)A$$

$$n(\lambda) = A + \frac{B}{\lambda^2} \leftarrow \text{cauchy's relation}$$

$$A.D = \frac{dS}{d\lambda} \propto \frac{1}{\lambda^3}$$





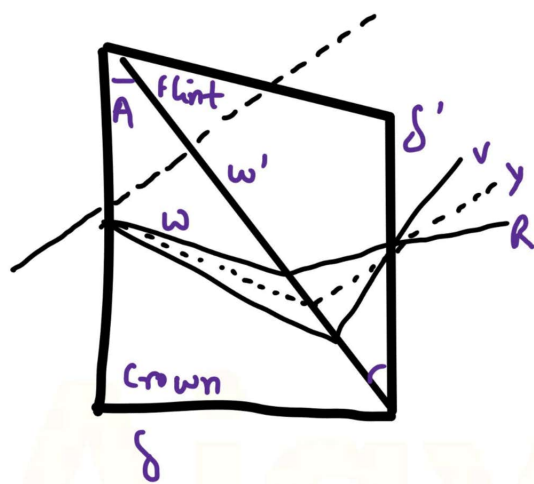
Angular Dispersion

$$\delta_v - \delta_R = (n_v - n_R) A$$

Dispersive Power

$$\frac{\delta_v - \delta_R}{\delta} = \frac{n_v - n_R}{n - 1}$$

Dispersion without Deviation (Achromatism)



$$\frac{A}{A'} = - \left(\frac{\mu'_v - \mu'_R}{\mu_v - \mu_R} \right)$$

Condition for Achromatism

$$\omega \delta + \omega' \delta' = 0$$

Chromatic Aberration

$$-\frac{df}{f} = \frac{dn}{n-1} \quad ; \quad -df = \omega f$$

for deviation,

$$D = \delta \left[1 - \frac{\omega}{\omega'} \right]$$

Optical Instruments:

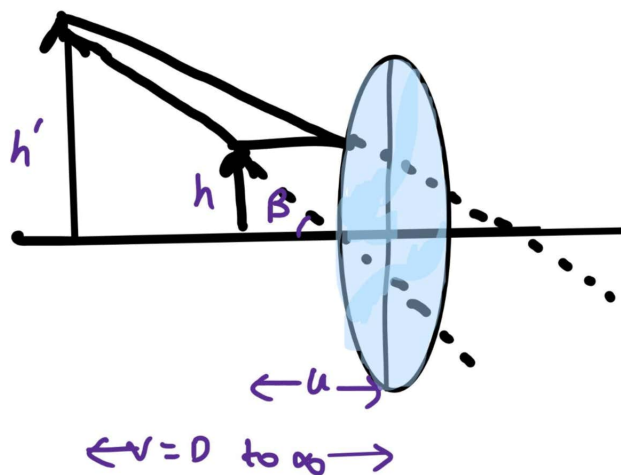
Angular Magnification:

$$m = \frac{\tan \beta}{\tan \alpha}$$





Simple microscope:



Magnifying Power

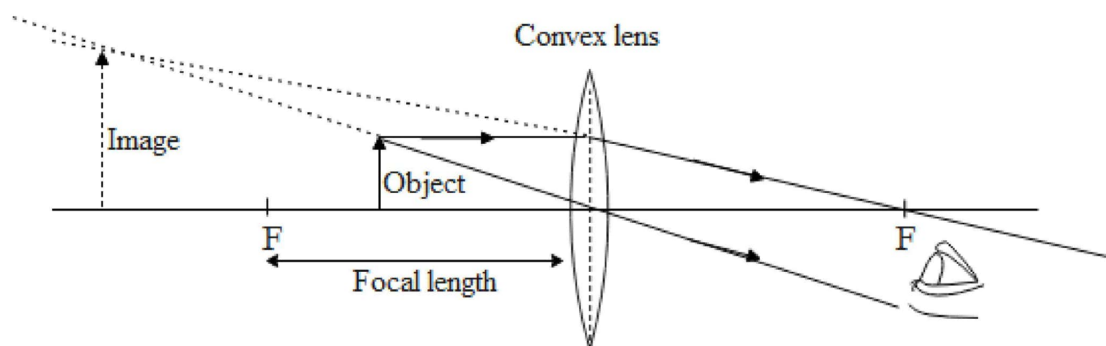
$$m = \frac{h'}{v} \times \frac{d}{h}$$

if $v = D$

$$y_{\max} = \frac{h'}{h} = 1 + \frac{D}{f}$$

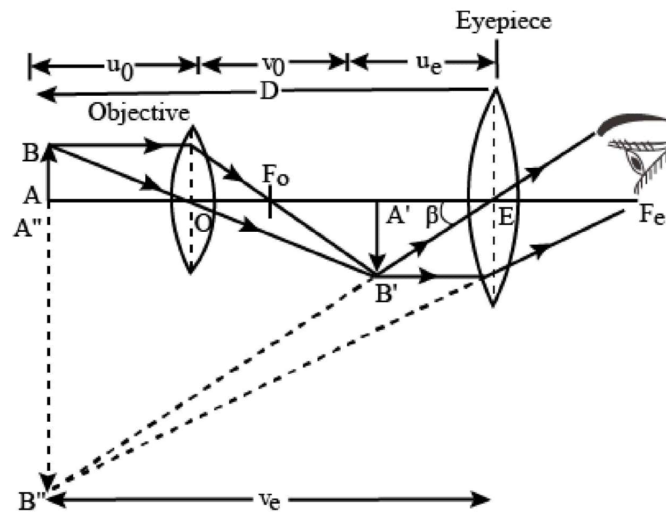
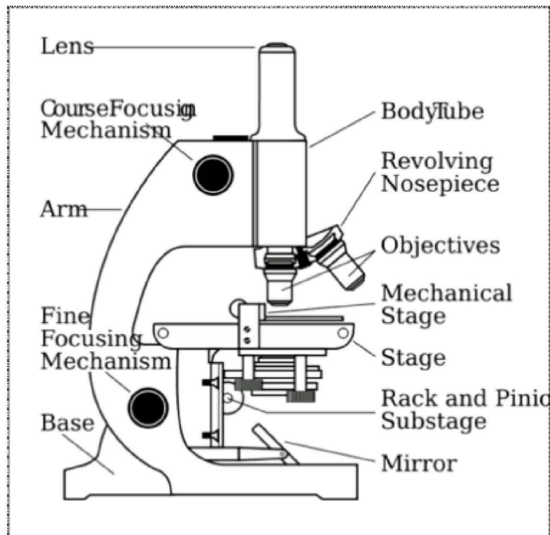
if $v = \infty$

$$y_{\min} = \frac{D}{f}$$





Compound Microscope



$$y = \frac{h'}{h} \times \frac{D}{u_e} = m y_e = \frac{-v}{u} y_e$$

when image is formed at $D =$ distance of clear vision of eye is strained

$$y = -\frac{v}{u} \left[1 + \frac{D}{f_e} \right]$$

Resolving power

$$\frac{1}{\Delta x} \propto \frac{1}{\lambda}$$

Δx is limit of resolution





Never miss any study material



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UNIT 9

WAVE OPTICSInterferenceIntensity (I_1)

$$y_1 = A_1 \sin \omega t$$

Intensity (I_2)

$$y_2 = A_2 \sin(\omega t + \phi)$$

Resultant Wave

$$y_R = A_R \sin(\omega t + \theta)$$

Resultant Amplitude

$$A = \sqrt{a^2 + b^2 + 2ab \cos \phi}$$

Resultant Intensities

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

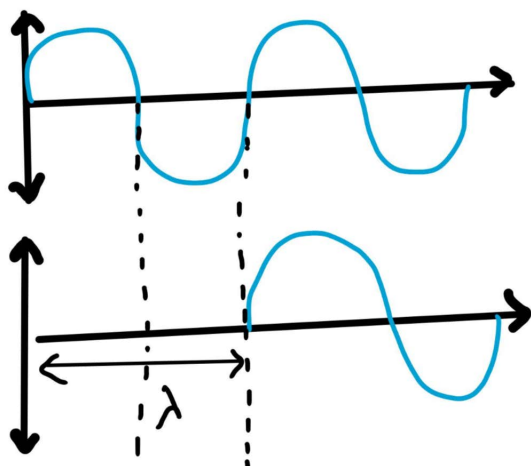
Relation b/w phase diff.

& path diff.

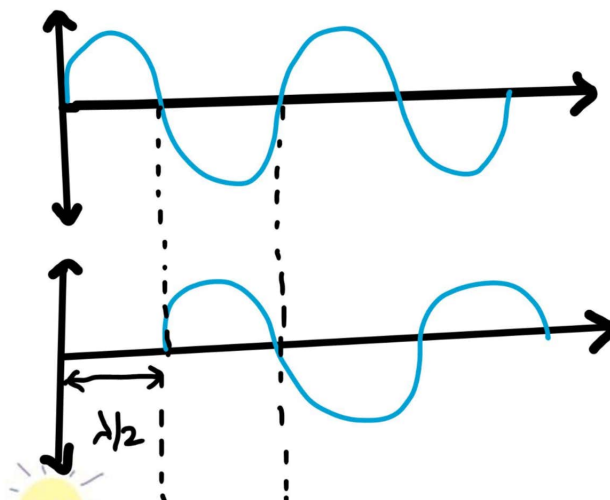
$$\Delta \phi = \frac{2\pi \Delta x}{\lambda}$$

Two types of interferences

Constructive



Destructive





Constructive

Path diff.

$$\Delta x = 0, \lambda, 2\lambda, \dots, n\lambda$$

Phase diff.

$$\Delta \phi = 2n\pi$$

$$n = 0, 1, 2, \dots$$

Amplitude

$$A_R = A_1 + A_2$$

Intensity

$$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

Time Interval

$$\theta = 2n\left(\frac{T}{2}\right)$$

Fringe width

$$\beta = \frac{\lambda D}{d}$$

Bright fringes

$$x = \frac{n\lambda D}{d}$$

Bright fringe

$$I_{\max} = 4I_0$$

Max Intensity when two same intensity sources are incident at a point.

Destructive

$$\Delta x = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \dots, (2n-1)\frac{\lambda}{2}$$

$$\Delta \phi = (2n-1)\pi$$

$$n = 1, 2, 3, \dots$$

$$A_R = A_1 - A_2$$

$$I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

$$\theta = (2n-1)\left(\frac{T}{2}\right)$$

$$\beta = \frac{\lambda D}{d}$$

Dark fringes

$$x = \frac{(2n-1)\lambda D}{2d}$$

Dark fringe

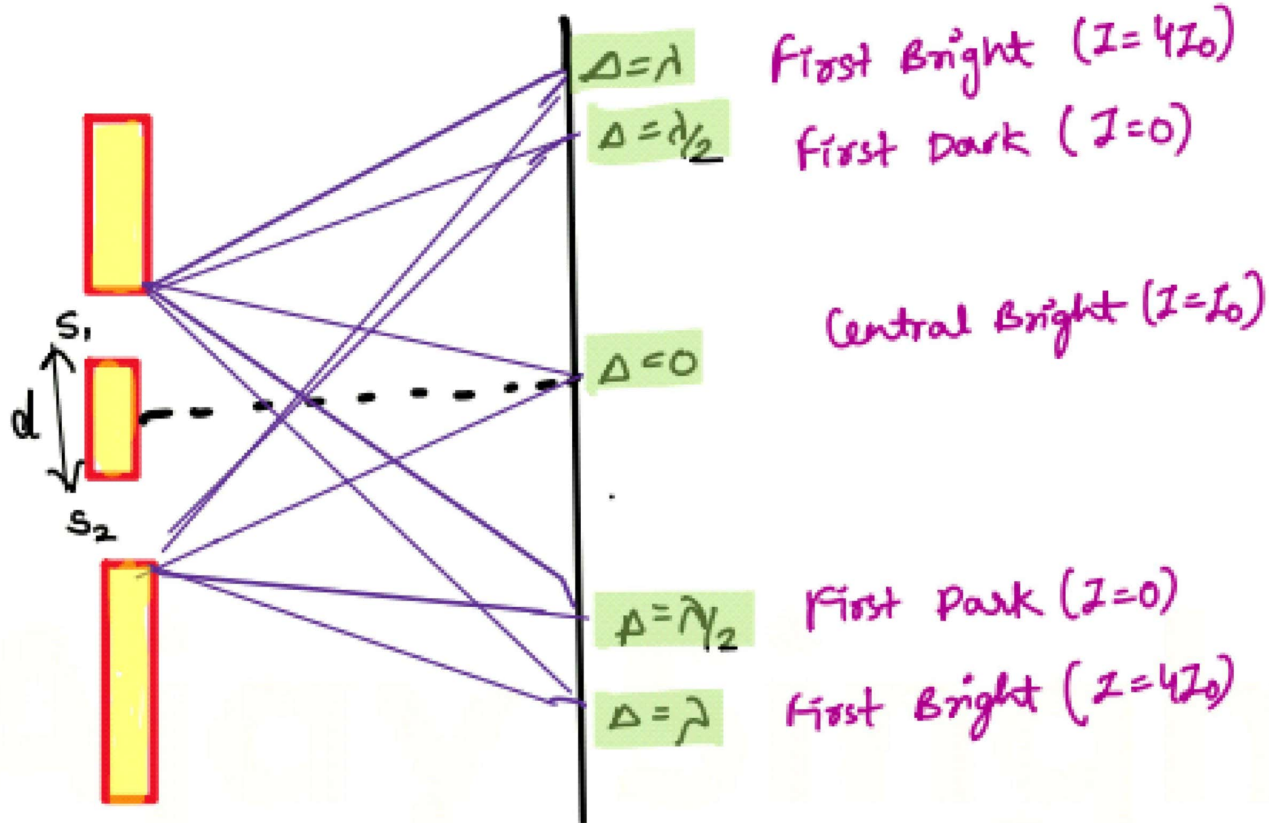
$$I_{\min} = 0$$

$$I_{\max} = 4I_0 \cos^2 \frac{\phi}{2}$$





Young's Double Slit Experiment



1) Path Difference

$$\Delta x = \frac{dx}{D}$$

2) Phase Difference

$$\Delta \phi = \frac{2\pi}{\lambda} \frac{dx}{D}$$

3) Intensity at any point
x distance from center

$$I_R = 4I_0 \left(\cos^2 \frac{\phi}{2} \right)$$

4) Bright fringe

$$x = \frac{n\lambda D}{d}$$

5) Dark fringe

$$x = \frac{(2n-1)\lambda D}{2d}$$





6) Fringe width

$$\beta = \frac{\lambda D}{d}$$

7) ROSE dipped in liquid

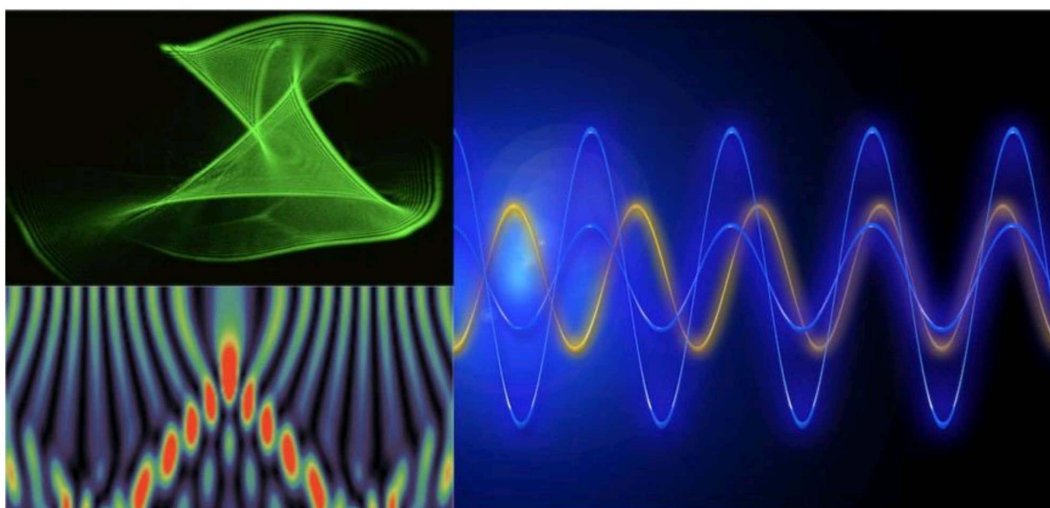
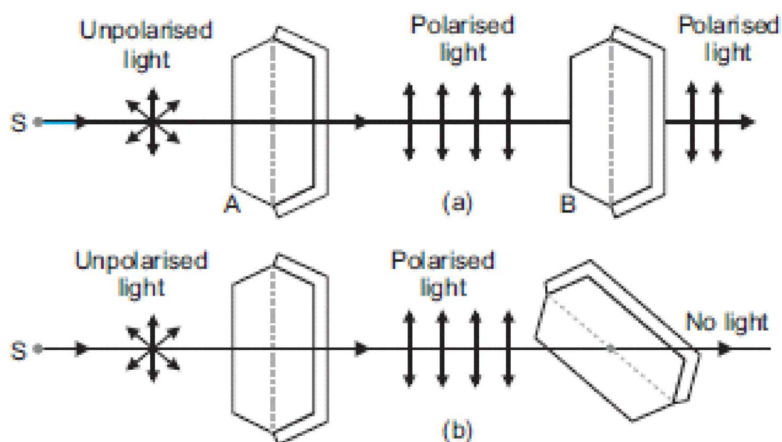
$$\lambda_m = \frac{\lambda_0}{n}$$

Malu's Law

$$I_i = k A_0^2$$

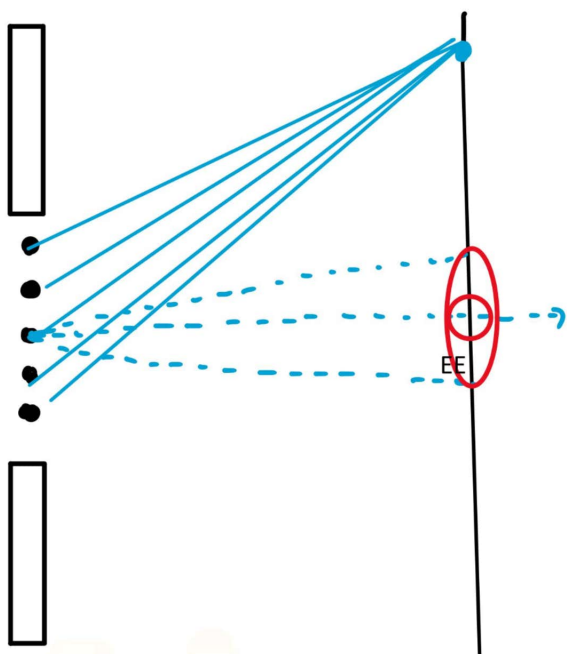
$$\langle I_t \rangle = \frac{I_0}{2}$$

$$I_t = I_i \cos^2 \theta$$





Diffraction



$$I = I_0 \frac{\sin^2 \beta}{\beta^2}$$

$$\beta = \frac{\pi b \sin \theta}{\lambda}$$

Minima

$$\sin \theta = \frac{n\lambda}{b}$$

$$\theta_{10} = \frac{\lambda}{b}$$

$$\theta_{20} = \frac{2\lambda}{b}$$

Maxima

$$\sin \theta = (2n-1) \frac{\lambda}{2}$$

$$\text{diameter} = 2\theta_{10} D$$

Bright Fringe

$$\Delta x = (2n-1) \frac{\lambda}{2}$$

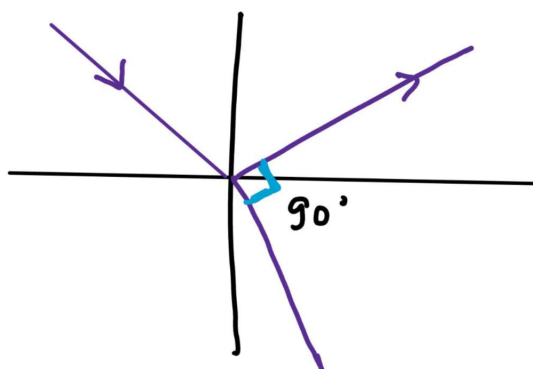
$$\Delta \phi = (2n-1)\pi$$

Dark fringe

$$\Delta x = n\lambda$$

$$\Delta \phi = 2n\pi$$

Brewster's Law



Brewster's Angle

$$\theta_B = \tan^{-1} n$$

Critical Angle

$$\theta_c = \sin^{-1} \frac{1}{n}$$





DUAL NATURE OF MATTER

Electron Emission

Thermionic Emission

$$J = AT^2 e^{-\phi_0/k_B T}$$

Field Emission

$$J = aE^2 e^{b|E|}$$

Photoelectric Effect

Einstein's Photoelectric Eq.

$$(E_k)_{\max} = h\nu - \phi_0 = [h\nu - \nu_0]$$

Stopping Potential

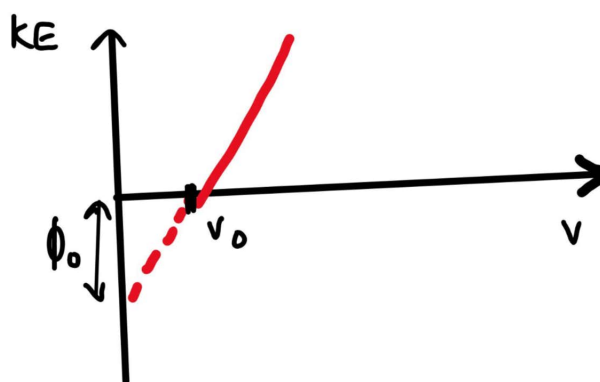
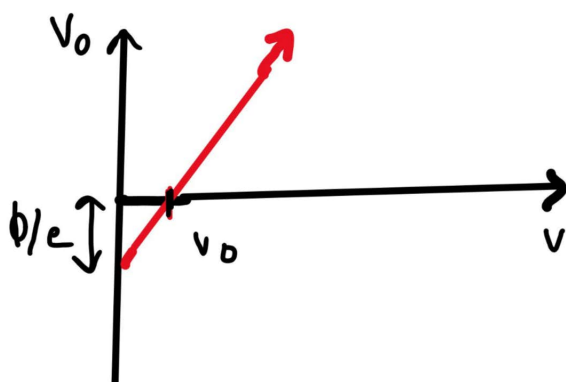
$$eV_0 = hc \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$$

Photoelectric efficiency

$$\frac{\text{no. of } e^- \text{ emitted}}{\text{no. of photons absorbed}}$$

Power

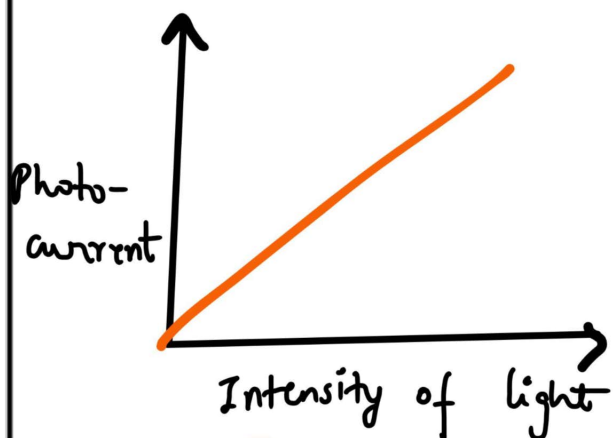
$$\text{Power} = n_p \epsilon_p$$



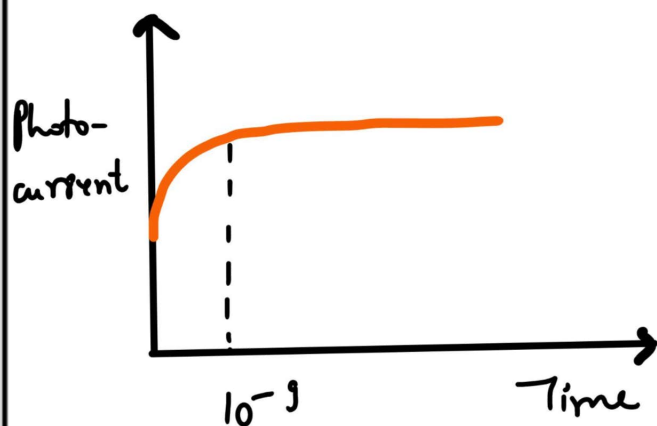


Important Graphs

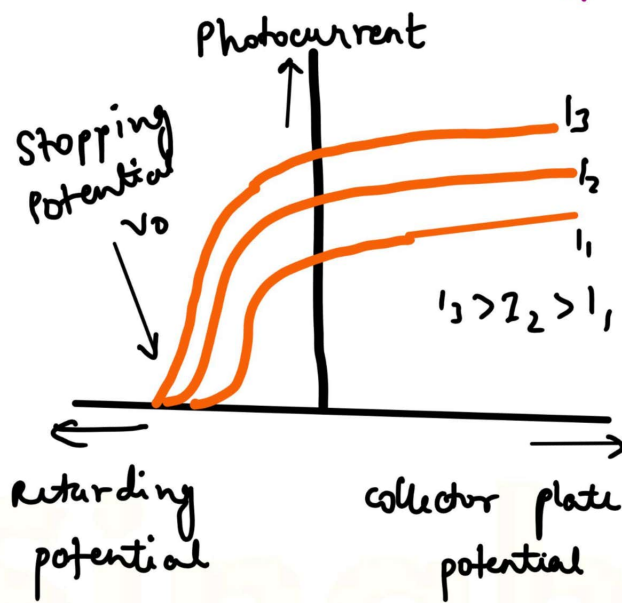
Variation of Photoelectric Current with intensity



Photocurrent v/s time

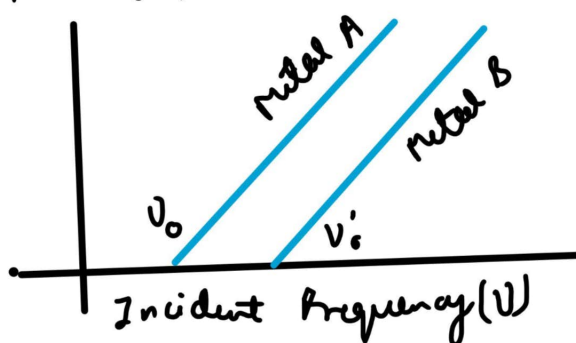


Variation of Photoelectric Current with potential diff.



V_0 v/s ν

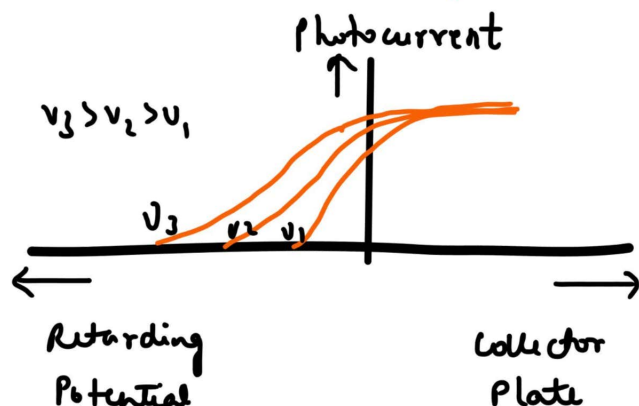
Stopping Potential (V_0)



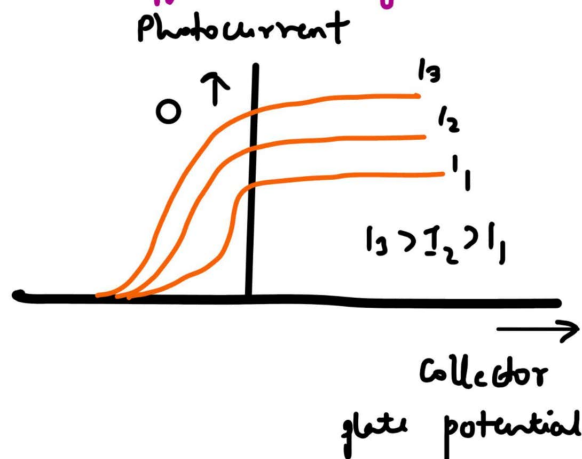


Photocurrent v/s voltage

with diff frequency



with diff. Intensity



De Broglie Equation:

Particle of mass m moving with velocity v .

particle in electric field

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

$$\lambda = \frac{h}{\sqrt{2m \text{ K.E.}}}$$

$$\lambda = \frac{h}{\sqrt{2mq \Delta V}}$$

Particle is moving in potential diff.

for electron

$$\lambda = \frac{12.2}{\sqrt{V}}$$

for proton

$$\lambda = \frac{0.286}{\sqrt{V}}$$

for deuteron

$$\lambda = \frac{0.202}{\sqrt{V}}$$

for α particle

$$\lambda = \frac{0.101}{\sqrt{V}}$$

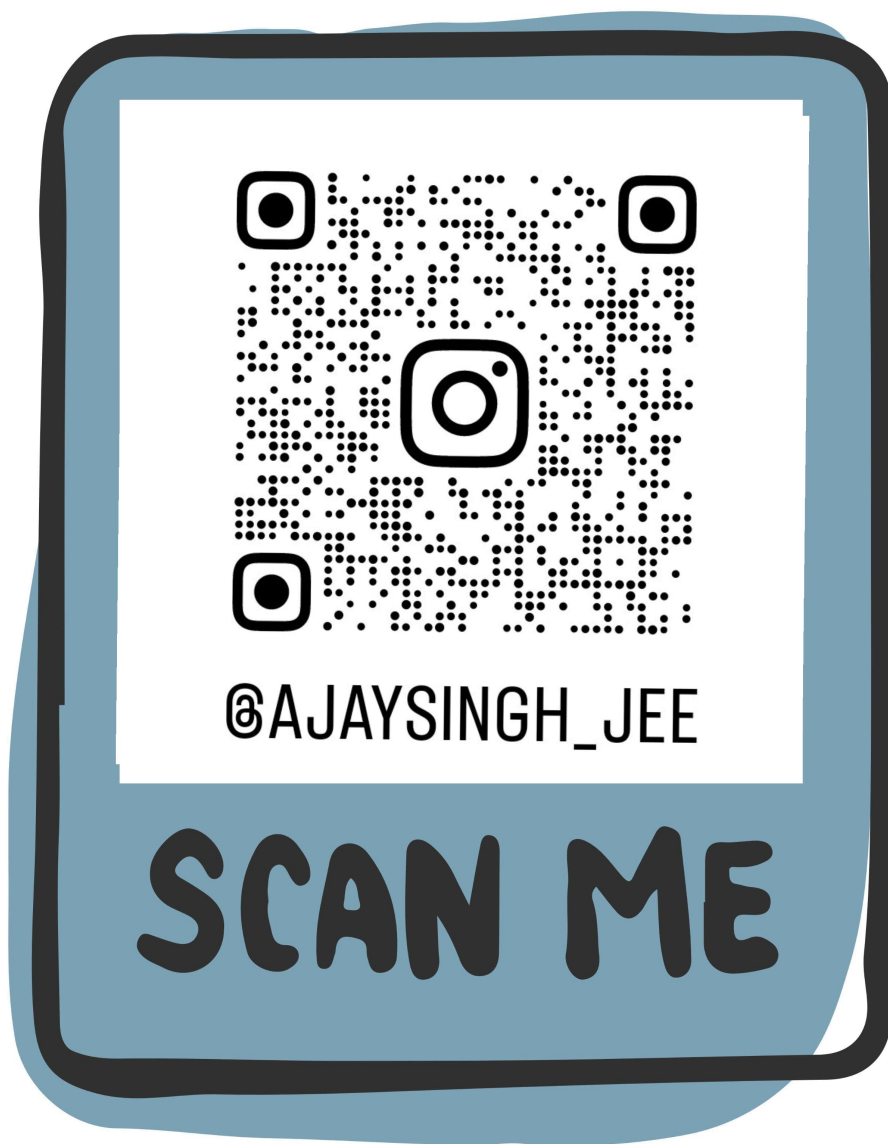
Collision

$$v_1 = \frac{(m_1 - em_2)u_1 + (1+e)m_2u_2}{m_1 + m_2} ; v_2 = \frac{(m_2 - em_1)u_2 + (1+e)m_1u_1}{m_1 + m_2}$$





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in magnetic field

$$\lambda = \frac{h}{qBr}$$

kinetic Theory of gases:

$$\lambda = \frac{h}{\sqrt{f m k_B T}}$$

$$\lambda_{\text{gas molecule}} = \frac{h}{\sqrt{3 m k_B T}}$$

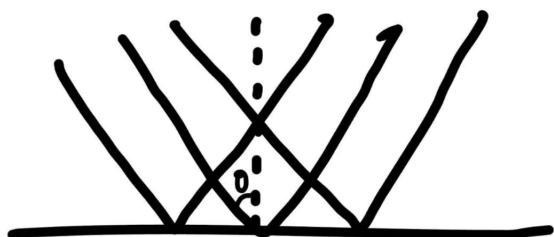
Heisenberg Uncertainty Principle:

It is impossible to predict simultaneously the position and momentum of an elementary particle.

$$\Delta x \Delta p \geq \frac{h}{2\pi}$$

Radiation and Pressure

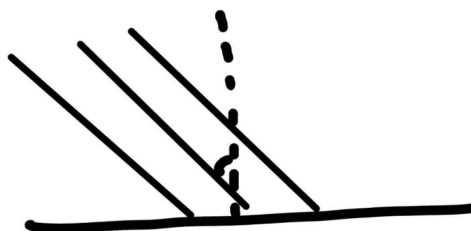
Surface with 100% reflection



$$F_{\text{plate}} = \frac{2IA \cos^2 \theta}{c}$$

$$P = \frac{2I}{c}$$

Surface with 100% absorption (Black body)



$$F_y = \frac{IA \cos^2 \theta}{c} \quad F_x = \frac{IA \sin \theta \cos \theta}{c}$$

$$P = \frac{I}{c}$$





ATOM

Rutherford Gold Foil Experiment



Application

Closest Distance of Approach

$$r_0 = \frac{2kZe^2}{E}$$

Impact Parameter

$$\cot \frac{\theta}{2} = \frac{2b}{r_0}$$

Bohr Atomic Model

Force

$$F_e = \frac{mv^2}{r}$$

$$\frac{kZe^2}{r^2} = \frac{mv}{r}$$

Angular
Momentum

$$L = mvr$$

$$mvr = \frac{nh}{2\pi}$$

Energy of
Electron

$$h\nu = |E_i - E_f|$$





Important Results

- 1) Radius of n^{th} orbit $r_n = \frac{a_0 n^2}{Z}$
- 2) Velocity of n^{th} orbit $v_n = \frac{v_0 Z}{n}$
- 3) Energy of n^{th} orbit $E_n = -13.6 \frac{Z^2}{n^2}$
- 4) Time period $T \propto \frac{n^3}{Z^2}$ $T = \frac{2\pi r_n}{v_n}$
- 5) Frequency of n^{th} orbit $\nu \propto \frac{1}{T} \propto \frac{Z^2}{n^3}$
- 6) Current in n^{th} orbit $I \propto \frac{Z^2}{n^3}$
- 7) Magnetic field $B = \frac{\mu_0 I n}{2r_n}$
- 8) Magnetic Moment $M \propto \frac{Z}{n}$
- 9) Area $A \propto \frac{n^4}{Z^2}$
- 10) Spectral series $\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$





NUCLEI

Important Points:

1) Size of Nucleus

$$R = R_0 A^{1/3}$$

2) Volume of Nucleus

$$V = \frac{4}{3} \pi R_0^3 A$$

3) Density of Nucleus

$$D = \frac{\mu}{\frac{4}{3} \pi R_0^3}$$

4) Mass Defect

$$\Delta m = [Z m_p + (A-Z) m_n] - M$$

5) Binding Energy

$$B.E. = \{ [Z m_p + (A-Z) m_n] - M \} c^2$$

6) Binding Energy per Nucleon

$$\frac{B.E.}{A} = \frac{\{ [Z m_p + (A-Z) m_n] - M \} c^2}{A}$$

7) Nuclear Stability

$$\frac{N}{Z} > 1.51$$

8) Q value of Nuclear Reaction

$$Q = (m_R - m_P) c^2$$

9) Nuclear Force

$$F_a \propto \frac{1}{r^7}, \quad F_Y = \frac{1}{r^8}$$





Radioactivity:

1) Rate of Disintegration

$$-\frac{dN}{dt} = \lambda dt$$

2) Number of nuclei present undecayed at any instant

$$N = N_0 e^{-\lambda t}$$

3) Half life

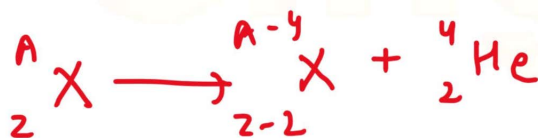
$$t_{1/2} = \frac{0.693}{\lambda}$$

4) Mean Life / Average Life

$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^{t/t_{1/2}}$$

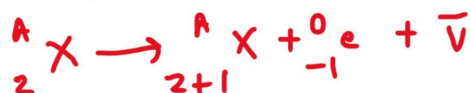
Types of Decay

1) Alpha Decay



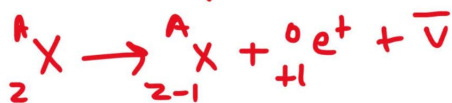
$$Q = [m_x - (m_y + m_{\text{He}})]c^2$$

2) Beta Decay Minus β decay



$$\Delta E = (M_x - M_y)c^2$$

plus β decay



$$\Delta E = (M_x - M_y - 2m_e)c^2$$

3) Gamma Decay





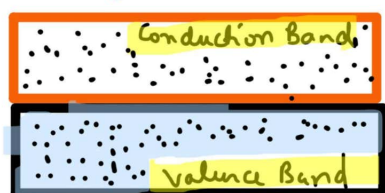
UNIT 12

SEMICONDUCTORS

Classification of solids according to Energy Band Theory

Conductors (Metals)

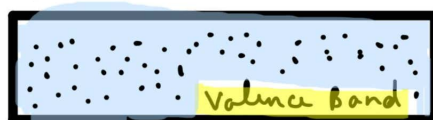
Overlapping Bands
 $E_g = 0$



Insulators

Empty Conduction Band

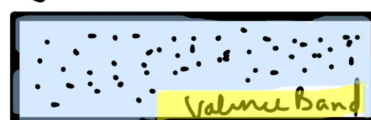
$E_g > 3\text{eV}$



Semiconductors

Conduction Band

$E_g < 3\text{eV}$

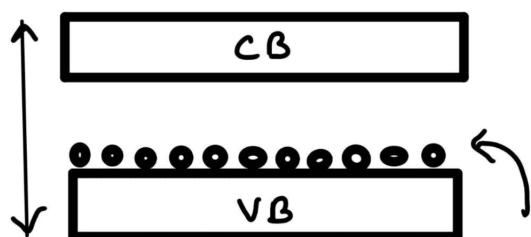


Classification of Semi Conductors:

Extrinsic Semiconductors

P-Type

Trivalent impurity is added
(B, In, Al)



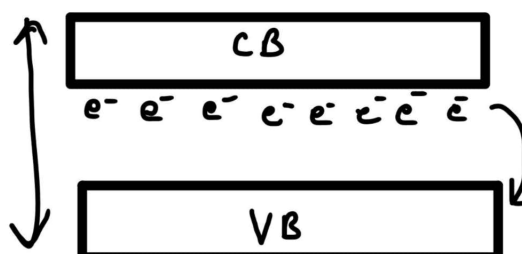
$$n_h \gg n_e$$

$$J \cong e n_h v_h$$

$$\sigma = \frac{1}{\rho} \cong e n_h \mu_h$$

N-Type

Pentavalent impurity is added
(P, As, Sb)



$$n_n \ll n_e$$

$$J \cong e n_e v_e$$

$$\sigma = \frac{1}{\rho} \cong e n_e \mu_e$$





Intrinsic Semiconductors

$$\text{Si} \rightarrow E_g = 1.1 \text{ eV}$$

$$\text{Ge} \rightarrow E_g = 0.68 \text{ eV}$$

$$n_e = n_h = n_i$$

$$J = ne[V_e + V_h]$$

Conductivity of
semiconductors

$$\sigma = e(n_e \mu_e + n_h \mu_h)$$

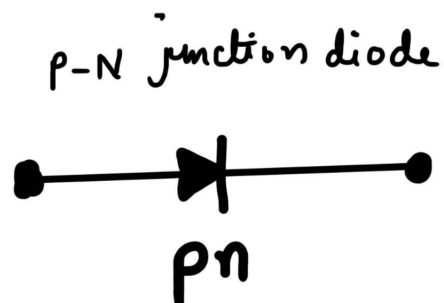
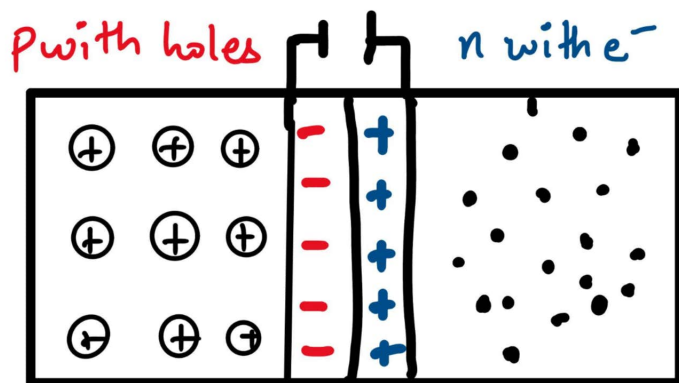
$$\sigma = \frac{1}{\rho} = en[\mu_e + \mu_h]$$

Mass Action Law

$$n_i^2 = n_h n_e$$

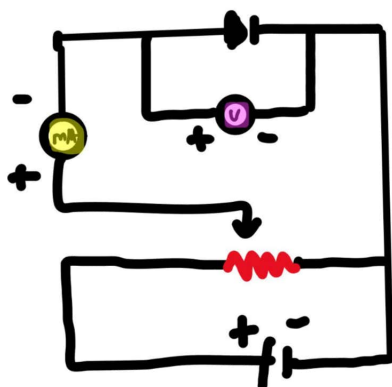
n_i^2 = intrinsic carrier density

P-N Junction Diode or Semiconductor Diode





forward Biasing



p end $\rightarrow$ connected with higher potential

n end $\rightarrow$ connected with lower potential

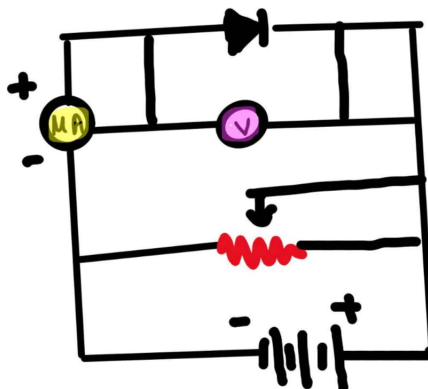
depletion layer decreases

$$V_p > V_n$$

$$V_p - V_n = \text{positive}$$

forward current flows in circuit

Reverse Biasing



p end $\rightarrow$ connected with lower potential

n end $\rightarrow$ connected with higher potential

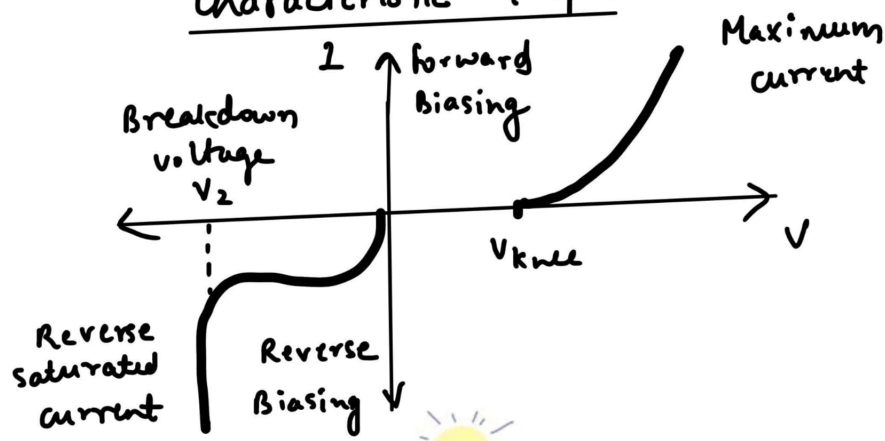
depletion layer increases

$$V_p < V_n$$

$$V_p - V_n = \text{negative}$$

Reverse current flows in circuit.

Characteristic Graph





Behaviour in Circuit:

Ideal Diode

Forward Bias	Reverse Bias
$R \rightarrow 0$	$R \rightarrow \infty$
Short circuit	Open circuit
$V_{knee} = 0$	$V_{knee} = 0$
$I = \frac{V_{ext}}{R_{ext}}$	No current will flow

Real Diode

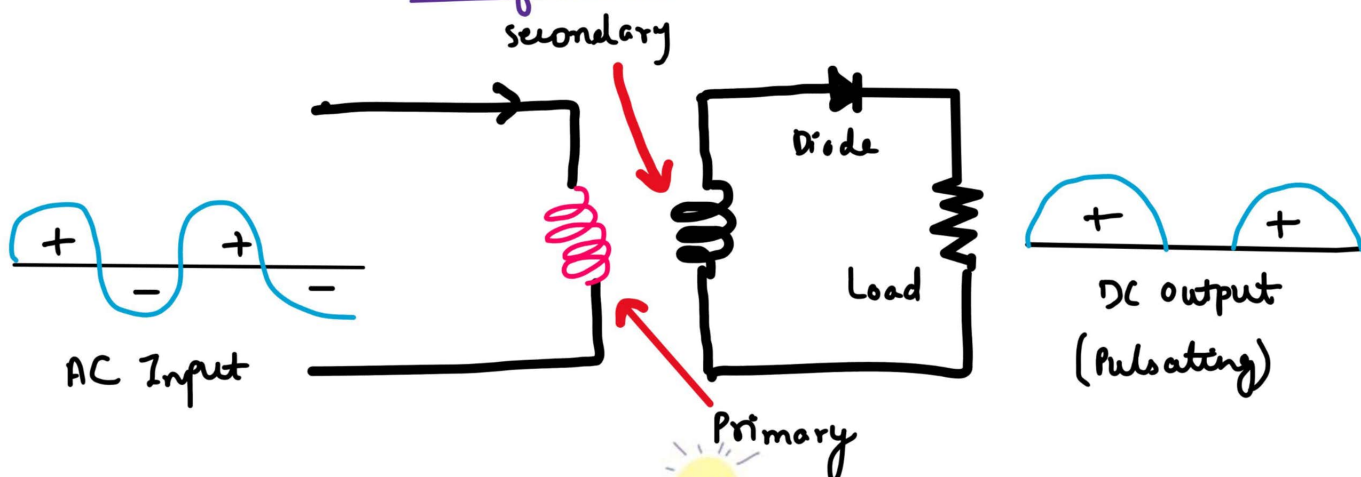
$$Si \rightarrow V_{knee} = 0.7V, Ge \rightarrow V_{knee} = 0.3$$

Forward Bias	Reverse Bias
Current flow from p to n	Current flow from n to p
$I = \frac{V_{ext} - V_{knee}}{R_{ext} + R_f}$	No current passed

Application of p-n junction Diode:

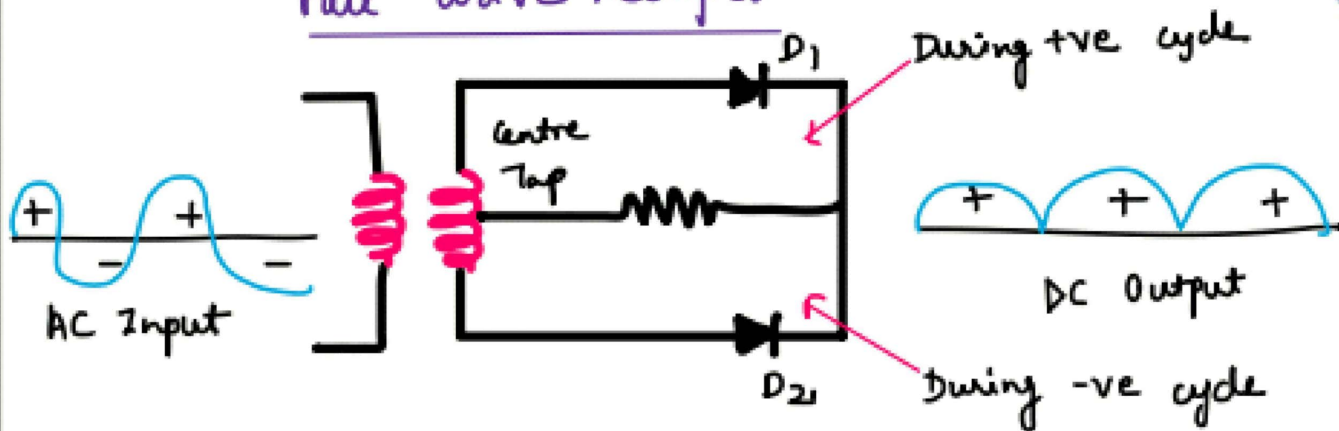
Rectifier — Converts AC input to DC output

Half wave Rectifier





Full Wave Rectifier



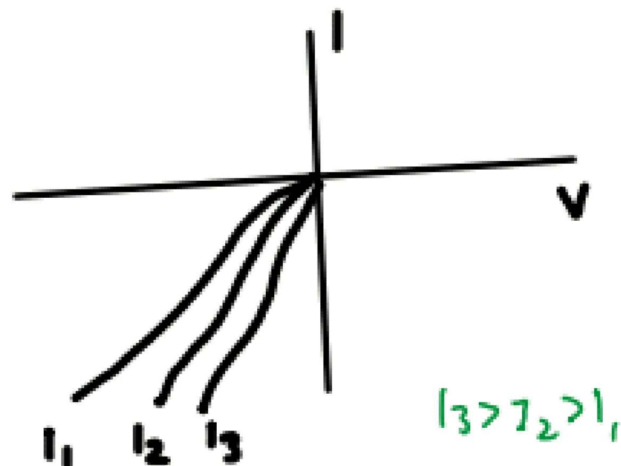
Light Emitting Diode

Less Power
Consumption)
Power losses)
heat losses

$$\lambda_{\text{emit}} = \frac{12400}{E_g} \text{ Å} \approx 100 \text{ Å} - 500 \text{ Å}$$

Photo Diode

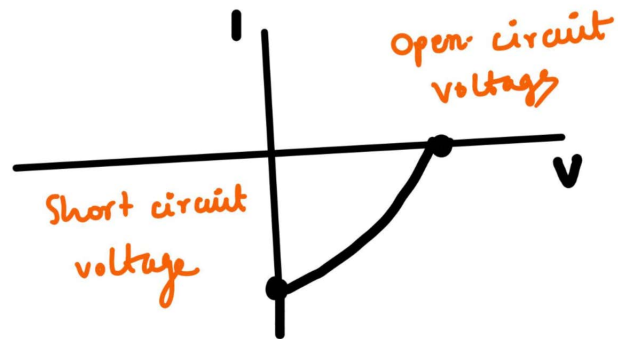
- Normal diode in reverse biased
- Used for detection of light





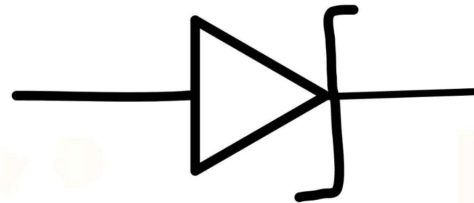
Solar cell

No biasing



Zener Diode

It is used as voltage regulator when used as reverse biased.



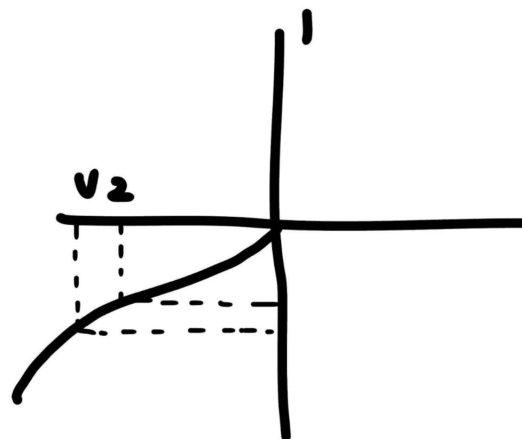
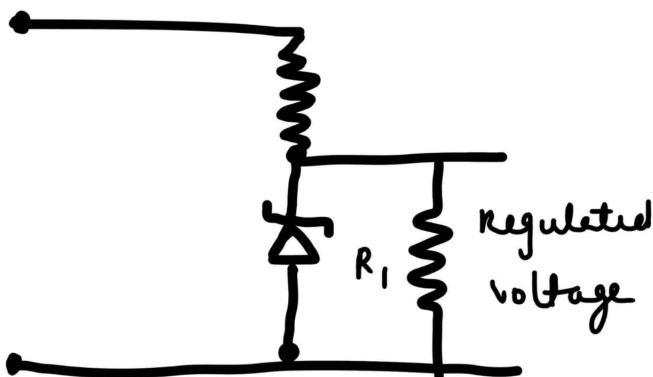
$$\text{Case (i)} = V_{\text{ext}} < V_Z$$

Behave as real diode

$$\text{Case (ii)} = V_{\text{ext}} \geq V_Z$$

V_Z will be fixed voltage

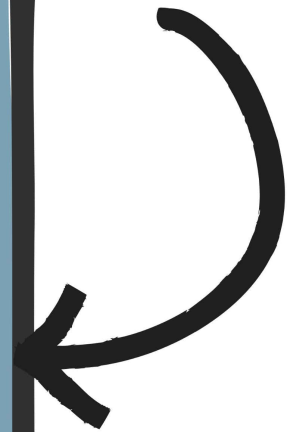
$$P_Z = V_Z I_Z$$



Every free resource update
here!!



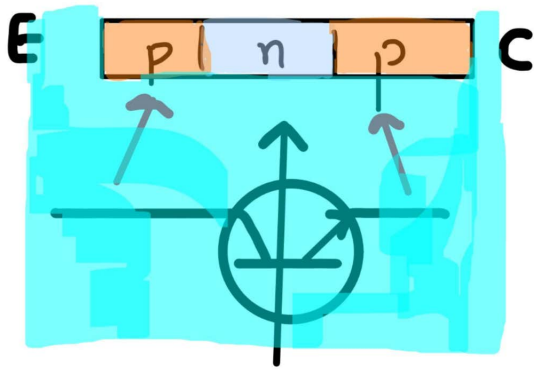
JOIN
US!



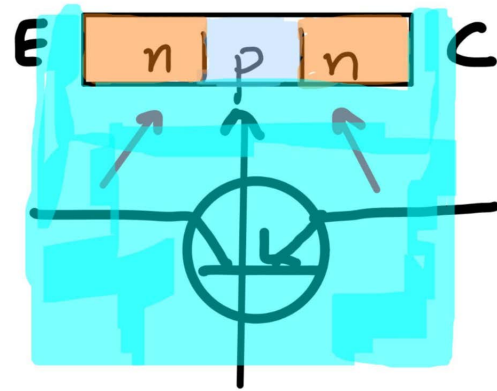


Transistors

Two junctions, Three terminal diodes.



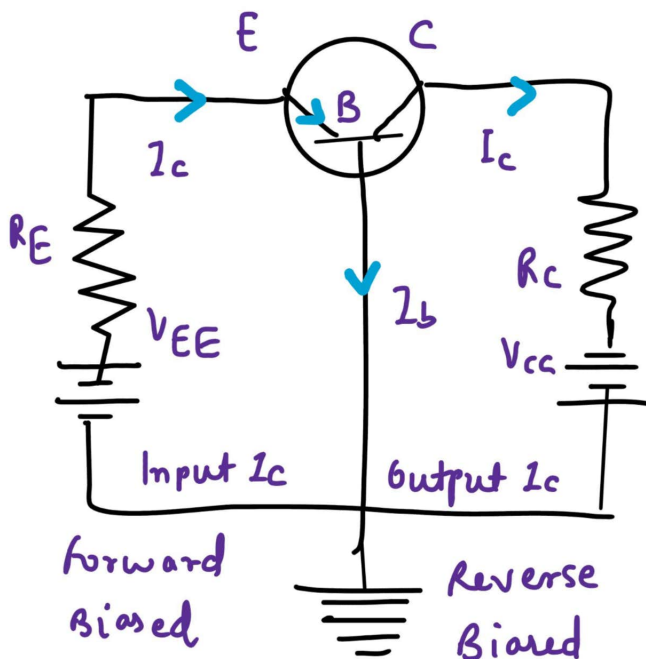
$$I_E = I_b + I_C$$



$$I_b \cong 5\% I_E$$

$$I_C \cong 95\% I_E$$

Common Base



Current Gain $\alpha = \frac{I_C}{I_E}$

Voltage Gain $\propto \frac{R_C}{R_E}$

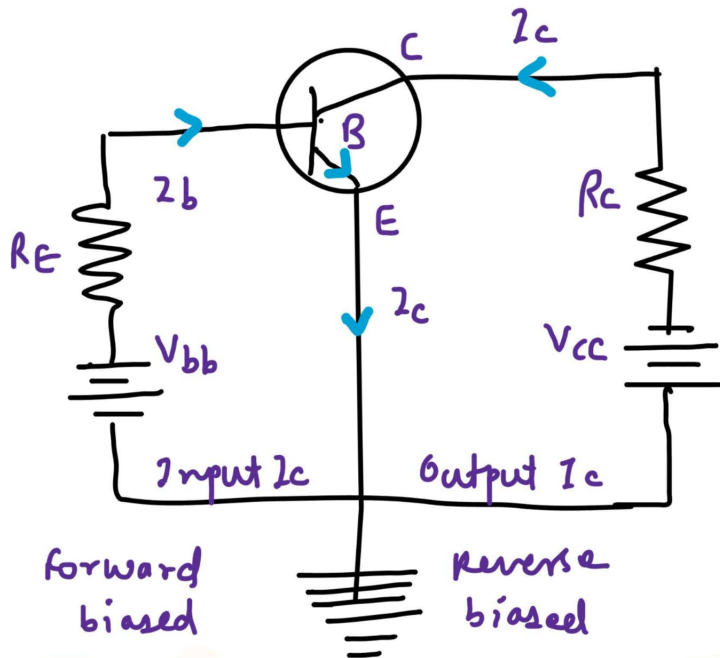
Power Gain $\propto \frac{R_C}{R_E}$

Phase Same phase





Common Emitter



Current Gain $\beta = \frac{I_C}{I_B}$

Voltage Gain $\beta \frac{R_C}{R_B}$

Power Gain $\beta^2 \frac{R_C}{R_B}$

Phase

Opp. phase

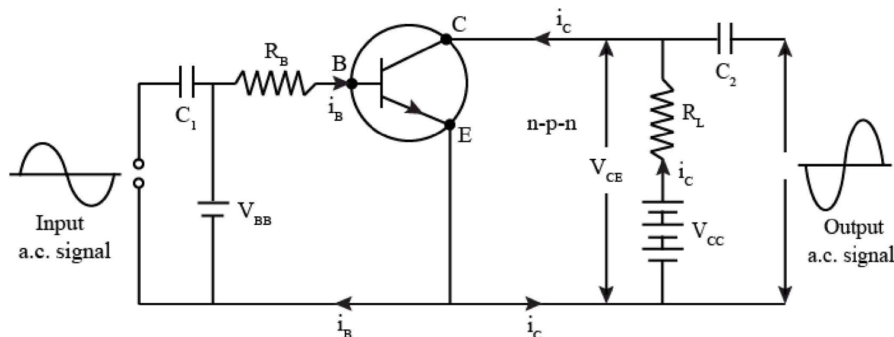
Relation b/w α and β

$$\alpha = \frac{\beta}{\beta + 1}$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

Transistor as a device

As an amplifier





with increase in input voltage (beyond 0.6 V)
 I_C increases and V_o decreases. (Active state)

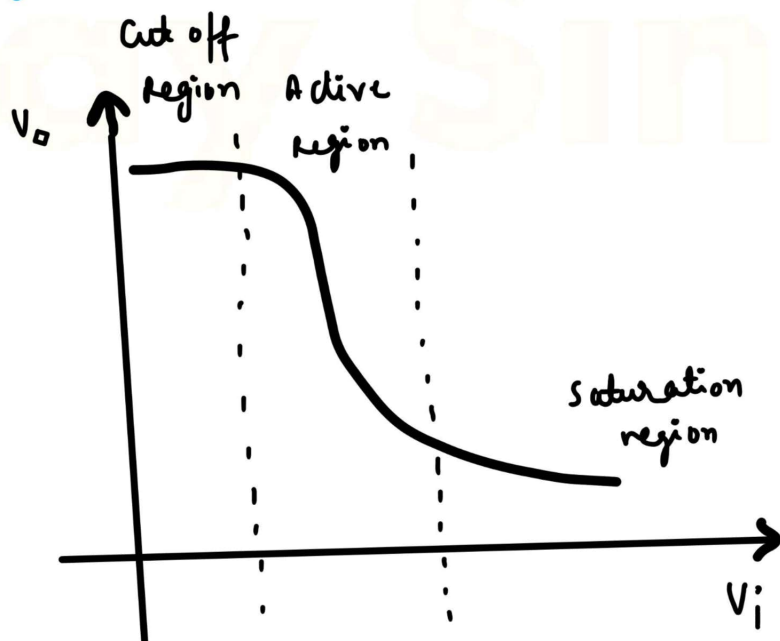
As a Switch:

with increase in input voltage ($0 - 0.6\text{ V}$)

$V_o = V_{CC} \text{ max}$ (off state)

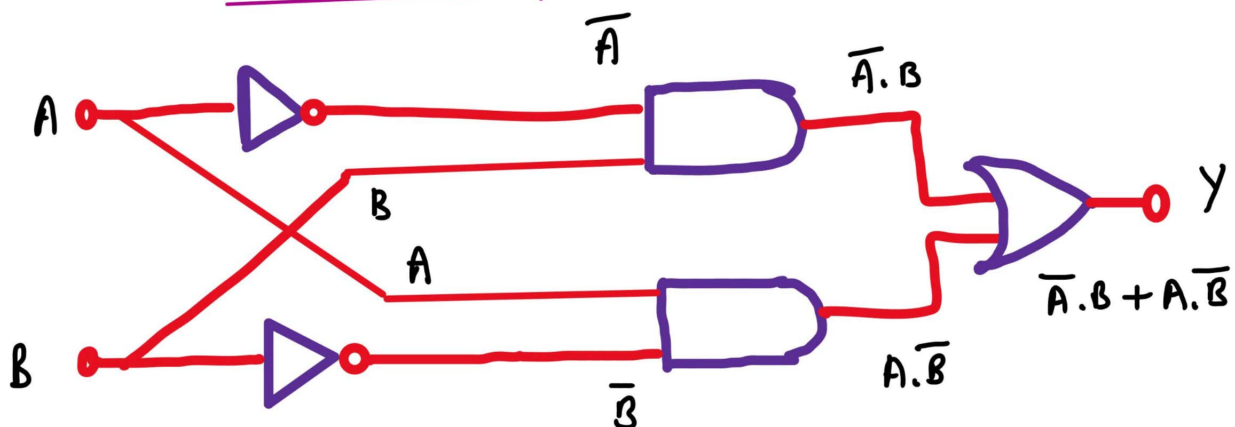
Non conducting transistor

I_C becomes max, output voltage decreases slowly (ON) stage.



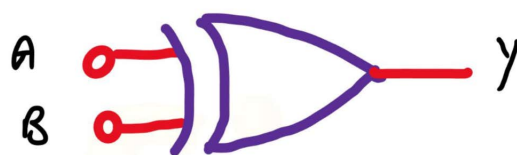


Exclusive-OR/XOR



$$Y = \bar{A}.B + A.\bar{B}$$

Symbol



A

B

Y

0

0

0

0

1

1

1

0

1

1

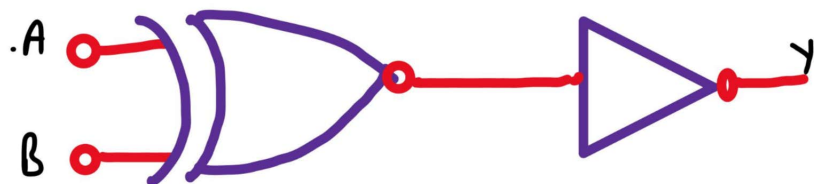
1

0

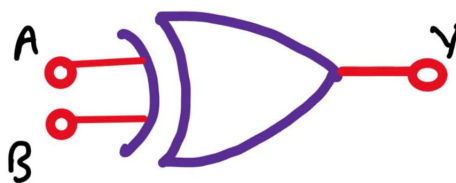




Exclusive - NOR

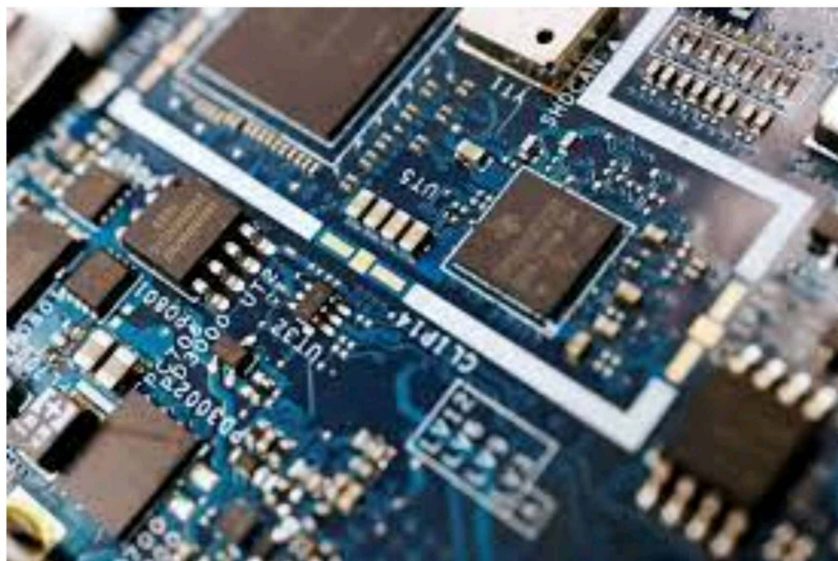


$$Y = \overline{A \cdot B + A \cdot \overline{B}}$$



A	0	0
B	0	1
Y	1	0

0	1
0	1
0	1





DE MORGAN LAW

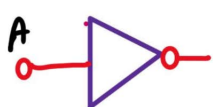
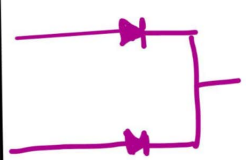
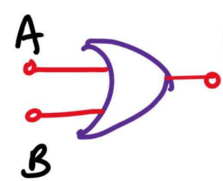
$$\overline{A+B} = \overline{A} \cdot \overline{B}$$

$$\overline{A \cdot B} = \overline{A} + \overline{B}$$

$$\overline{\overline{A+B}} = A+B$$

Different Combinations

Input		OR	NOR	AND	NAND	X-OR	X-NOR
A	B						
0	0	0	1	0	1	0	1
0	1	1	0	0	1	1	0
1	0	1	0	0	1	1	0
1	1	1	0	1	0	0	1

Gate	Boolean Expression	formed Using	circuit symbol	Electrical circuit
NOT	$Y = \overline{A}$	CE Transistor		key parallel with bulb
OR	$Y = A+B$			Two keys in and in series with bulb

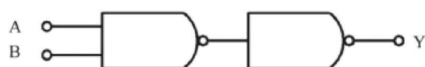




NOR	$Y = \overline{A+B}$ $= \overline{A} \cdot \overline{B}$	Diode + CE Transistor		Two keys in and in parallel with bulb
AND	$Y = A \cdot B$			Two keys in series and in series with bulb
NAND	$Y = \overline{A \cdot B}$ $= \overline{A} + \overline{B}$	CE + Two diode		Two keys in series and in parallel with bulb.

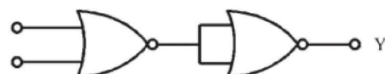
Questions to try

35 Which gate will be obtained by combination of following gates ?



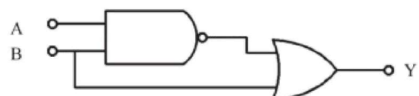
- (A) XOR (B) AND (C) NAND (D) OR

36 Which gate will be obtained by combination of following gates ?



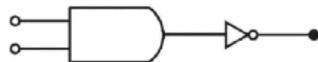
- (A) NOT (B) NAND (C) XOR (D) OR

37 Which of following is correct option for given combination of gates ?



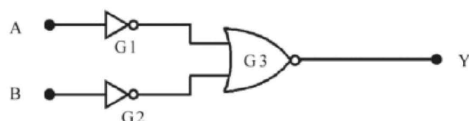
- (A) $Y = \overline{A} \cdot B + \overline{B}$ (B) $Y = \overline{A} \cdot B + \overline{B} \cdot \overline{A}$ (C) $Y = \overline{A} \cdot B + B$ (D) $Y = (\overline{A} + \overline{B}) \cdot \overline{B}$

38 For the given combination of gate which option from the following is correct ?



- (A) $\overline{A+B} = X$ (B) $A+B = X$ (C) $A \cdot B = X$ (D) $\overline{A+B} = X$

39 Following combination of gates will behave as which gate ?



- (A) AND gate (B) XOR gate (C) NOR gate (D) NAND gate



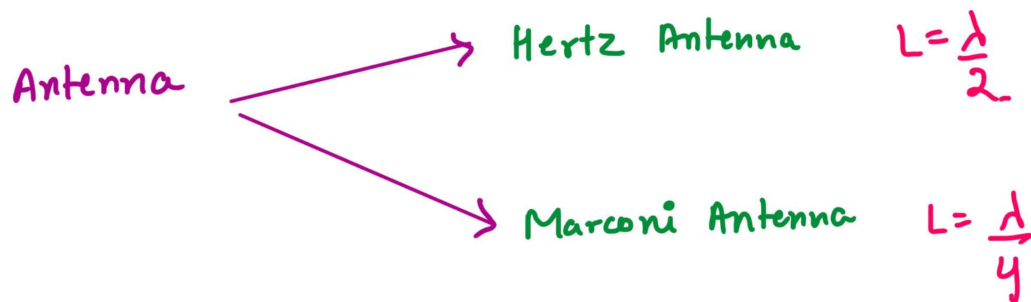


UNIT 13

COMMUNICATION

Band width of signal
(range of frequency)

$$BW = 2 \times n \times HF$$



Type of wave:

Ground wave - This is to propagate signals with lower frequencies

Sky wave -

- frequency 30-40 MHz
- long distance communication through ionosphere

$$V_{max} = 9\sqrt{D}$$

Space wave -

- frequency > 40 MHz
- direct transmitter → receiver
- for LOS communication

$$D_t = \sqrt{2Rh_T} \quad D_M = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

Modulation

$$m(t) = A_m \sin \omega_m t \quad c(t) = A_c \sin \omega_c t$$

Final wave - $A_c \left[1 + \left(\frac{A_m}{A_c} \right) \sin \omega_m t \right] \sin \omega_c t$

$$\mu = \frac{A_m}{A_c}$$





Love You All ..

~AJAY SINGH

**Good luck!
Seize the day and
give it everything
you've got.**

